

1. Administrative & Study Identification

Full Study Title: A Phase 3, Multicenter, Double-blind, Randomized, Placebo-controlled Clinical Study to Evaluate the Safety and Efficacy of Ertugliflozin (MK-8835/PF-04971729) in Pediatric Participants (Ages 10 to 17 Years, Inclusive) With Type 2 Diabetes Mellitus **Short Title/Acronym:** Ertugliflozin Type 2 Diabetes Mellitus (T2DM) Pediatric Study (MK-8835/PF-04971729) (MK-8835-059) **Protocol Number:** MK-8835-059 **NCT Number:** NCT04029480 **Sponsor Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Ertugliflozin (MK-8835) **Phase of Development:** Phase 3 **Medical Monitor:** Sponsor Medical Monitor (Merck Sharp & Dohme LLC)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether the addition of ertugliflozin reduces hemoglobin A1C (HbA1C) more than the addition of placebo after 24 weeks of treatment in pediatric participants with T2DM on metformin with or without insulin. **Secondary Objectives:** To evaluate the safety and tolerability of ertugliflozin in a pediatric population and assess long-term glycemic control up to 54 weeks.

2.2 Background & Rationale Type 2 diabetes in youth is a growing global health concern characterized by an increasing incidence. Pediatric-onset T2DM is often more aggressive and leads to earlier, more severe long-term cardio-renal complications compared with adult-onset T2DM. With limited approved therapeutic options that effectively slow beta-cell deterioration, this study is conducted to investigate whether the sodium-glucose co-transporter 2 (SGLT2) inhibitor ertugliflozin can provide a clinically meaningful improvement in glycemic control as an add-on therapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (parallel-group assignment) **Trial Status:** Completed

3.2 Planned Enrollment & Duration Target \$N\$: 166 participants **Number of Sites:** Multicenter **Estimated Study Start:** 08-Oct-2019 **Estimated Primary Completion:** 10-Apr-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Be ≥ 10 years and ≤ 17 years of age at the time informed consent is signed. 2. Has diabetes diagnosed by one of the American Diabetes Association (ADA) criteria. 3. Has a body mass index (BMI) ≥ 85 th percentile at screening OR participant has a history of being overweight or obese at the time of diagnosis of Type 2 diabetes mellitus (T2DM). 4. T2DM for ≥ 2 years, OR T2DM for < 2 years and a fasting C-peptide value > 0.6 ng/mL at Screening. 5. On stable metformin monotherapy (≥ 1500 mg/day) for ≥ 8 weeks prior to Screening, OR on a stable metformin dose (≥ 1500 mg/day) for ≥ 8 weeks prior to Screening and a stable dose of insulin for ≥ 8 weeks prior to Screening. 6. Contraceptive use by male participants should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies. 7. Is a non-sterilized female who is currently not sexually active OR who agrees to abstain from heterosexual activity OR who agrees to start contraception prior to initiating sexual activity and who agrees to use an adequate method of contraception (consistent with local regulations). 8. Have a family member or adult who, along with the participant, will be closely involved in the participant's daily activities (in the opinion of the investigator) and in the participant's treatment and study procedures.

4.2 Exclusion Criteria 1. Has known type 1 diabetes mellitus or documented evidence of positive diabetes autoantibodies performed when the participant was diagnosed with diabetes. 2. Has known monogenic diabetes, or secondary diabetes. 3. Has symptomatic hyperglycemia and/or moderate to large ketonuria requiring immediate initiation of another antihyperglycemic agent, including insulin. 4. Has a known hypersensitivity or intolerance to any sodium glucose co-transporter 2 (SGLT2) inhibitor. 5. Is pregnant, or breast feeding or is expecting to conceive or donate eggs during the study, including 14 days following the last dose of study medication. 6. Has previously taken an SGLT2 inhibitor (such as canagliflozin, dapagliflozin, empagliflozin, or ertugliflozin) or was enrolled in a study for these agents. 7. Has a history of idiopathic acute pancreatitis or chronic pancreatitis. 8. Has a history of severe hypoglycemia while on insulin.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants are initially randomized on Day 1 to receive either 5 mg ertugliflozin (ERTU) (and placebo to 15 mg ERTU) or matching placebo (placebo to 5 mg ERTU and placebo to 15 mg ERTU). At Week 12, participants who meet up-titration criteria are re-randomized for Weeks 12 to 54 to: - 5 mg ERTU and placebo to 15 mg ERTU (5 mg/5 mg Ertugliflozin) - 15 mg ERTU and placebo to 5 mg ERTU (5

mg/15 mg Ertugliflozin) Participants who do not meet up-titration criteria remain on 5 mg ERTU (and placebo to 15 mg) from Week 12 to Week 54. The placebo arm continues receiving placebo from Week 12 to Week 54. **Route of Administration:** Oral (Tablets). **Duration of Treatment:** 54 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background therapy of metformin with or without insulin. Initiation of investigator-discretion rescue therapy (including insulin) is permitted if strict glycemic rescue criteria are met. **Prohibited:** Initiation of other antihyperglycemic agents outside of predefined glycemic rescue protocols.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent/ Assent, Medical History, BMI, baseline HbA1c/C-peptide
Baseline	Day 0	Randomization, First Dose of ERTU or Placebo
Interim 1	Week 12	Safety Labs, Efficacy Assessment, Up-titration evaluation/Re-randomization
Interim 2	Week 24	Primary Efficacy Assessment (HbA1C), Safety check
End of Study	Week 54	Final Vital Signs, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Change From Baseline in Hemoglobin A1C (HbA1C) at Week 24 (Combined Ertugliflozin Versus Placebo):** Determined via whole blood samples (percentage of glycated HbA1C). A negative number indicates a reduction in A1C level. Analyzed using Bayesian mean change from baseline for the combined ertugliflozin arms versus placebo.
- 2. Number of Participants Who Experienced an Adverse Event (AE) Up to Week 24:** An AE is defined as any unfavourable and unintended sign, symptom, or disease temporally associated with the use of a medicinal product or protocol specified procedure.
- 3. Number of Participants Who Experienced an AE Up to Week 54:** Evaluated for the combined ertugliflozin arms versus placebo throughout the entire 54-week study duration.

7.2 Secondary Outcomes

- 1. Change From Baseline in Hemoglobin A1C at Week 24 (Dose-optimized Ertugliflozin Versus Placebo)**
- 2. Change From Baseline in Hemoglobin A1C at Week 24 (5 mg Ertugliflozin Versus Placebo)**
- 3. Change From Baseline in Fasting Plasma Glucose (FPG) at Week 24**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical monitoring via patient reporting and laboratory analysis. Specific monitoring for known SGLT2i risks such as genital mycotic infections, diabetic ketoacidosis (DKA), and hypoglycemia. Any worsening of a preexisting condition temporally associated with the sponsor's product is recorded as an AE. **Serious Adverse Events (SAEs):** Required reporting within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External independent safety committee evaluating unblinded safety signals throughout the trial duration.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all AE/SAE evaluations. Ertugliflozin arms are combined for the primary efficacy and primary safety (AE) analyses. **Sample Size Justification:** A sample of 166 participants provides adequate statistical power to detect meaningful differences in the primary estimand using posterior credible intervals. **Handling of Missing Data:** Rubin's rule multiple imputation methodology is utilized to manage missing data for the primary efficacy analysis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Risk-based monitoring featuring periodic onsite verification of source data, coupled with remote clinical data review. **Audit Trail:** Validated eCRF systems with comprehensive audit trails for data entry, locking, and data clarification forms (DCFs).

11. Study Identifiers & Governance

Primary ID: MK-8835-059 **Registry Number:** NCT04029480 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** Ertugliflozin Type 2 Diabetes Mellitus (T2DM) Pediatric Study (MK-8835/PF-04971729) **Official Regulatory Title:** A Phase 3, Multicenter, Double-blind, Randomized, Placebo-controlled Clinical Study to Evaluate the Safety and Efficacy of Ertugliflozin (MK-8835/PF-04971729) in Pediatric Participants (Ages 10 to 17 Years, Inclusive) With Type 2 Diabetes Mellitus

12. Clinical Framework (Pediatric T2DM Specific)

Primary Condition: Type 2 Diabetes Mellitus (T2DM) in Pediatric Patients **Diagnosis Threshold:** ADA criteria diagnostic thresholds + BMI $\geq$ 85th percentile or history of overweight/obesity at diagnosis **Medical Methodology:** Addition of SGLT2 inhibitor to stable metformin $\pm$ insulin **Optimization Target:** Hemoglobin A1C (HbA1C) reduction at 24 weeks

13. Study Procedures & Methodology

Management Pathway: SGLT2 Inhibitor therapy pathway integrated with standard background anti-diabetic treatments. **Education Protocol (HCP):** Trial procedures, dosing up-titration criteria at Week 12, and glycemic rescue guidelines. **Education Protocol (Patient):** Counseling on diet, physical activity, and symptom recognition for hypoglycemia and ketoacidosis. **Quality Audit Mechanism:** Periodic site monitoring and central statistical data tracking.

14. Observation & Follow-Up Schedule

Total Duration: 54 treatment weeks **Onsite Visit Cadence:** Baseline, Week 12, Week 24, and Week 54 **Remote Monitoring Frequency:** Standard safety check-ins per clinical site protocols. **Checklist Review Interval:** Regular review of concomitant medications, background insulin dose stability, and potential adverse events.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				